

TOI-4087b

R-2165

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI4087_260607 · created 2026-08-13T00:05:43+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461198.742 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1336 +/- 0.0096
Transit depth (Rp/Rs)^2	1.78 +/- 0.26 [%]
Orbital Inclination (inc)	86.85 +/- 1.18
Airmass coefficient 1 (a1)	10.94 +/- 0.21
Airmass coefficient 2 (a2)	-1.5 +/- 0.62
Scatter in the residuals of the lightcurve fit is	1.86 %
Best Comparison Star	#2 - [396, 340]
Optimal Aperture	3.42
Optimal Annulus	12.29
Transit Duration (day)	0.0648 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1336 $\pm$ 0.0096 vs 0.10754068997360952 (deviation 1.13 σ)
Duration fit vs published	0.0648 d vs 0.1242 d (deviation 3.3 σ)
Mid-time O-C	36.3 min
Depth SNR	5.8
Residual scatter	1.86 %

Light curve

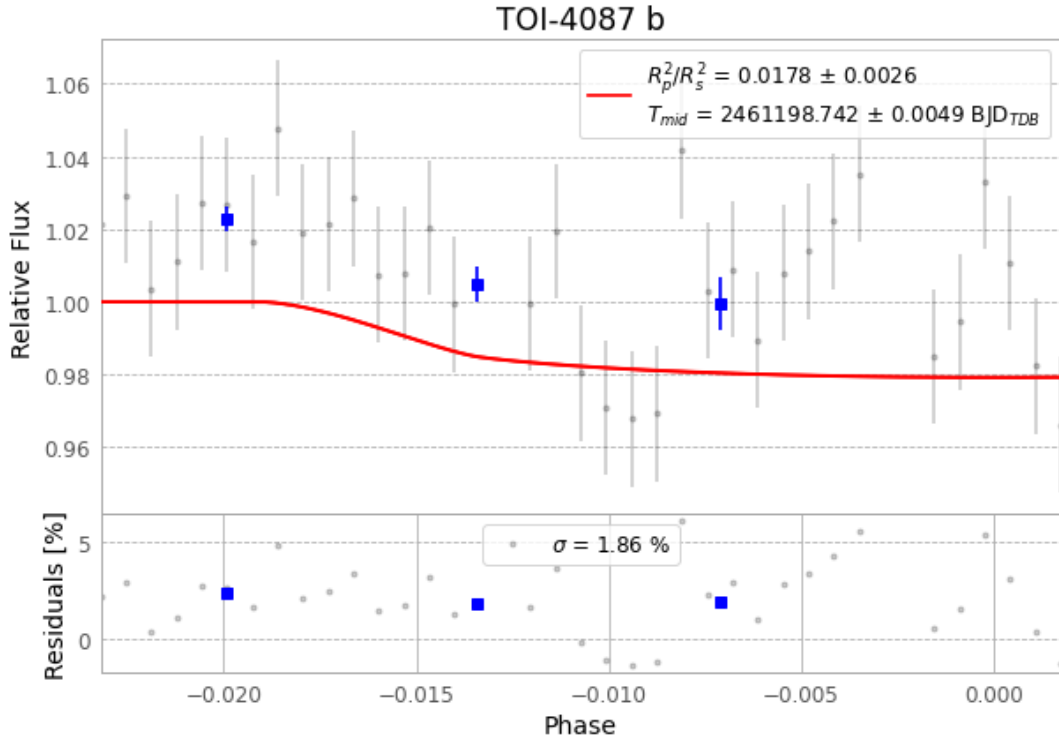


Figure 1 — FinalLightCurve_TOI-4087_b_2026-06-06.png (SHA-256 588bdd82274faa20...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [327, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d63c50b8fd57e4ca...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

39 FITS frames (26 MB), TOI4087260607040315.FITS → TOI4087260607055715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4087-260607.log (644341895d30...), FinalLightCurve_TOI-4087_b_2026-06-06.png (588bdd82274f...), FinalLightCurve_TOI-4087_b_2026-06-06.csv (ebb5aa4a713d...)

Compute resources

Wall time 477.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:05Z.